

Naomi (Yu-Shan) Cheng

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science GPA: 4.0

Champaign, IL

Expected Dec 2026

Honors: Tau Beta Pi (Engineering Honor Society), Phi Kappa Phi (Top 10% Graduate Students)

Relevant Coursework: Cloud Computing Applications; Machine Learning, Artificial Intelligence, Natural Language Processing, Practical Statistical Learning; Text Information Systems; Theory and Practice of Data Cleaning, Data Visualization

New York University

Diploma in Financial Planning

New York, NY

Sep 2018 – Dec 2020

PROFESSIONAL EXPERIENCE

International Business Machines (IBM)

Lowell, MA

Software Engineer Intern

May – Aug 2026

- Engineered a TypeScript/Playwright automation framework for IBM Guardium Data Security Center SaaS, co-designing a layered Spec–Service–Page architecture and authoring 2,300+ LOC of production-quality code refined through iterative peer review and adopted by three engineers.
- Developed reusable TypeScript API utilities and shared components for test-data setup and cleanup, supporting end-to-end automation of event-driven and custom-report workflows while reducing duplicated implementation across the codebase.
- Modernized 30 business-critical workflows by migrating them from Java/Selenium to TypeScript/Playwright and integrating automated regression execution into Jenkins CI, delivering 3× faster runtime.
- Improved framework reliability through state-based synchronization, targeted retry mechanisms, and transient UI recovery, reducing fixed waits by 64%.

IBM watsonx Hackathon — Guardium Adaptive Insights

- Collaborated with full-time IBM engineers as the team's only intern to prototype an agentic AI experience for IBM Guardium, integrating LLMs, RAG, MCP, and adaptive data visualization.
- Implemented multi-user personalization with Llama, translating natural-language preference updates into user-keyed JSON state and applying retrieved preferences to subsequent visualization rendering.

Educational Testing Service (ETS)

Princeton, NJ

Assessment Specialist

Nov 2020 – Sep 2025

AI-Powered Automation (Jun - Sep 2025)

- Contributed to automating ~95% of targeted exam-item generation workflows across ETS flagship exams (TOEIC, TOEFL, PRAXIS, GRE) by translating assessment requirements into standardized LLM prompt workflows through iterative experimentation and validation.

Test Development (2020 – 2025)

- Worked directly with College Board client stakeholders while leading an Agile team of 4 external SMEs, managing evolving requirements and delivery expectations to achieve ~90% committee acceptance and 100% on-time delivery, supporting contract renewal.

TECHNICAL PROJECTS

ExamFlux — AI-Powered Exam Authoring and Review Platform [\[GitHub | Live Demo\]](#)

Tools: Python, FastAPI, SQLite, React (JavaScript, HTML, CSS), OpenAI API, Docker

- Built an end-to-end full-stack AI exam-authoring platform with React and FastAPI, integrating LLM generation, human review and approval, REST APIs, and database persistence, support ~100 monthly reviewers.
- Designed a modular, object-oriented FastAPI backend with SQLAlchemy/SQLite persistence and structured routers, services, models, and schemas to support generation, review, approval, and database workflows.

Cloud Computing Applications — Distributed Systems & Infrastructure

Tools: Python, SQL, AWS, Kubernetes, Docker, Apache Spark, Apache Flink, Redis, MySQL

- Built end-to-end cloud and distributed systems workflows on AWS, implementing S3/Glue/PySpark ETL pipelines for data transformation, containerized ML deployment on EKS, and distributed batch/stream processing with Spark and Flink.
- Engineered supporting infrastructure including Kubernetes workload isolation and resource quotas, Aurora MySQL–Redis caching with lazy-loading/write-through strategies, and event-driven stream processing with Kinesis and Flink.

SKILLS

- Programming Languages: TypeScript, JavaScript, Python, SQL, Java, HTML/CSS
- Technologies: React, FastAPI, Playwright, Git, Docker, AWS(S3, Lambda, Glue, EKS, Aurora), MySQL, Redis